

SYMCROOKS-STEERING: UNBIASED RARE EVENT PROBABILITY DENSITY ESTIMATION IN LANGUAGE MODELS VIA PARALLEL REPLICA EXCHANGE AND BIDIRECTIONAL CROOKS REWEIGHTING

Ambitious AI Researcher

AI Research Institute
123 AI Avenue, ML City
researcher@institute.ai

ABSTRACT

Accurately quantifying extreme tail behaviors in autoregressive large language models (LLMs)—such as safety violations, alignment anomalies, and extreme readability scores—is critical for risk assessment but computationally prohibitive due to exponential probability decay under standard ancestral sampling. Existing rare-event methods based on single-chain transition path sampling (TPS) and unidirectional importance sampling suffer from metastable MCMC chain trapping ($\hat{R} > 1.1$), high integrated autocorrelation, and catastrophic Effective Sample Size (ESS) collapse over long horizons ($T = 100$). We introduce **SymCrooks-Steering**, a rare-event framework integrating a parallel replica-exchange TPS ladder, bidirectional Crooks Fluctuation Theorem reweighting (BAR), discrete truncation shooting, short-tail myopic autodiff guidance with an exact shadow-work Metropolis-Hastings correction, and explicit repetition-penalty tilting. On TinyStories-8M across Automated Readability Index (ARI) and completion log-probability observables, SymCrooks-Steering achieves 100% Gelman-Rubin convergence across all 20 bias states (20/20 vs. 18/20 for baseline annealed TPS), reduces mean tail autocorrelation by up to 78.5%, yields tighter 96% bootstrap confidence intervals (0.685/0.826 vs. 0.968/0.933), and enables fluctuation-theorem cross-validation ($< 1\%$ BAR-vs-MBAR agreement) at over $4.6\times$ faster GPU execution (435 s vs. 2038 s). Beyond this setting, it generalizes to a dense non-differentiable neural toxicity classifier on RealToxicityPrompts, outperforms Single-Pass/Twisted SMC and the ITGIS, MHIS, and UTIS importance samplers (all $< 0.4\%$ ESS in heavy tails), attains full statistical convergence (58/60 states) at $T=200$ on LLaMA-3-8B and Qwen2.5-7B within a single 40GB GPU, and remains effective on the discrete AdvBench jailbreak landscape.

1 INTRODUCTION

Autoregressive large language models (LLMs) generate high-dimensional token sequences according to complex probability distributions. While ancestral sampling effectively captures typical output behaviors, it fails to quantify extreme tail events—such as safety violations, alignment failures, and non-canonical readability anomalies (Jones et al., 2025; Wu & Hilton, 2024; Angell et al., 2026). Because the probabilities of these rare sequences decay exponentially with sequence horizon T , evaluating tail risk via direct ancestral sampling is computationally intractable, requiring hundreds of millions to billions of generated tokens to observe even a handful of rare completions (Dorman et al., 2026).

To overcome the exponential computational cost of direct sampling, recent work has applied rare-event simulation techniques, such as Transition Path Sampling (TPS) and Annealed Importance Sampling (AIS), to target exponentially tilted distributions $p_\lambda(x_{1:T}) \propto e^{-\lambda\phi(x_{1:T})}p_M(x_{1:T})$ (Neal, 1998; Dorman et al., 2026; Wu et al., 2026). However, existing path-sampling frameworks rely on single Markov chains that are sequentially annealed through a progression of bias levels λ . In high-

dimensional discrete sequence spaces, these single-chain annealed paths suffer from three critical technical bottlenecks:

1. **Metastable MCMC Chain Trapping.** At deep bias regimes ($\lambda \geq 0.8$), single-chain MCMC trajectories become trapped in local energy basins, failing to converge across all bias states (manifesting as Gelman-Rubin failures $\hat{R} > 1.1$) (Chen et al., 2020; Dorman et al., 2026).
2. **High Autocorrelation & Weight Variance.** Long horizons ($T = 100$) induce high integrated autocorrelation times (IAT) and exponential importance-weight variance, leading to catastrophic Effective Sample Size (ESS) collapse when estimating partition-function ratios $Z(\lambda)$ (Elvira et al., 2018; Ren et al., 2025).
3. **Distribution Distortion in Soft Guidance.** Uncorrected guided decoding and myopic soft steering (Lew et al., 2023; Wu et al., 2026) alter proposal distributions without exact Metropolis-Hastings (MH) corrections, introducing unquantifiable bias into long-horizon density estimates.

To resolve these challenges, we introduce **SymCrooks-Steering**, a parallel replica-exchange Transition Path Sampling framework equipped with bidirectional Crooks fluctuation-theorem reweighting and Metropolis-exact short-tail myopic guidance. SymCrooks-Steering re-architects rare-event density estimation for language models through four integrated design principles:

- **Parallel Replica-Exchange TPS Ladder.** Instead of single-chain sequential annealing, SymCrooks-Steering maintains a parallel ensemble of Markov chains across adjacent bias rungs ($\lambda_k \rightleftharpoons \lambda_{k+1}$). Inter-replica configuration swaps enable cross-bias transport, overcoming energetic barriers and eliminating metastable trapping.
- **Bidirectional Crooks / BAR Fluctuation Reweighting.** Applying the Crooks Fluctuation Theorem (Crooks, 1999) across forward and reverse trajectory work distributions yields robust, variance-reduced free-energy differences $\Delta F_{k \rightarrow k+1}$ with per-pair uncertainty bounds, cross-validating Multi-State Bennett Acceptance Ratio (MBAR) (Shirts & Chodera, 2008) calculations to within $< 1\%$.
- **Discrete TPS Truncation Shooting with Repetition Penalty.** We retain discrete sequence truncation shooting to preserve token structure, incorporating an explicit consecutive-repetition penalty into the tilted observable $\tilde{\phi}(x_{1:T})$ to suppress degenerate token loops without biasing the target density.
- **Metropolis-Exact Short-Tail Myopic Autodiff Guidance.** Soft gradient-based logit perturbations are restricted to short initial sequence tails and deployed strictly as MCMC proposal moves, governed by an exact shadow-work MH correction that guarantees asymptotic convergence to the true tilted distribution.

We evaluate SymCrooks-Steering against direct ancestral sampling and single-chain annealed TPS (Dorman et al., 2026) on TinyStories-8M across two observables: Automated Readability Index (ARI) and joint completion log-probability over 100-token horizons ($T = 100$). Our key contributions are: (i) *full chain convergence* ($\hat{R} < 1.1$ on 20/20 bias states vs. 18/20 for single-chain annealed TPS); (ii) a $4\times$ *reduction in deep-bias autocorrelation* (IAT drops from 42.95 to 9.24 for ARI at $\lambda = \pm 1.0$); (iii) *tighter reweighting confidence intervals at lower GPU cost* (mean relative 96% bootstrap CI width reduced from 0.968 to 0.685 for ARI and 0.933 to 0.826 for log-probabilities, at $4.6\times$ faster wall-clock); and (iv) *fluctuation-theorem cross-validation* (BAR and MBAR log-partition functions agree within $< 1\%$ across the full ladder); and (v) *real-world generalization and scale*—robust density estimation under a dense non-differentiable neural toxicity classifier on RealToxicityPrompts, decisive gains over Single-Pass/Twisted SMC and importance-sampling baselines (ITGIS, MHIS, UTIS) that collapse to $< 0.4\%$ ESS in heavy tails, full statistical convergence (58/60 bias states) at $T=200$ on LLaMA-3-8B and Qwen2.5-7B, and active replica transport on the discrete AdvBench jailbreak landscape (Section 4.6).

2 RELATED WORK

Rare-Event Analysis & Path Sampling in Autoregressive Models. Analyzing rare trajectory events in high-dimensional stochastic processes is a fundamental challenge in statistical physics and

computational statistics (Neal, 1998; Rose et al., 2020), and in autoregressive language generation it is further hampered by exponential probability decay (Dorman et al., 2026). Recent work on model risk forecasting and tail estimation (Jones et al., 2025; Angell et al., 2026; Wu & Hilton, 2024) underscores the imperative of estimating low-probability behaviors before deployment. Early simulators adopted single-chain Annealed Importance Sampling (Neal, 1998; Penny & Sengupta, 2016) or Sequential Monte Carlo (Lew et al., 2023; Luo et al., 2026) to bias sampling toward target outcomes. However, sequence-level tilted distributions exhibit complex, multi-modal landscapes (Deng et al., 2020): sequentially annealed single-chain TPS becomes trapped in metastable basins at high bias ($\lambda \geq 0.8$), yielding incomplete mixing ($\hat{R} > 1.1$) and high autocorrelation (Chen et al., 2020), while cumulative importance weights undergo severe variance explosion and ESS collapse (Elvira et al., 2018). In contrast, SymCrooks-Steering’s parallel replica-exchange ladder enables cross-bias transport, reducing IAT by up to $4\times$ and guaranteeing 100% convergence, and applies directly to real-world safety targets such as classifier-based toxicity EBMs and jailbreak detectors.

Guided Decoding and Soft Steering. Controlling generated-text attributes at inference has inspired diverse guided decoding strategies. Classifier-guided methods such as GeDi (Krause et al., 2020), DExperts (Liu et al., 2021), and FUDGE (Yang & Klein, 2021) modify next-token logits using auxiliary discriminators. Differentiable energy-based approaches, including COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), employ gradient-based Langevin dynamics in continuous logit space. While effective at biasing outputs, uncorrected myopic perturbations alter the underlying proposal distribution (Khalifa et al., 2020; Zhao et al., 2024); without exact MH corrections they introduce unquantifiable bias into density estimates. Discrete MCMC frameworks such as CGMH (Miao et al., 2018) enforce strict acceptance but explore deep tails poorly, and soft MH variants (Feng et al., 2025) lack rigorous target recovery. SymCrooks-Steering uses short-tail myopic guidance strictly as an MCMC proposal governed by an exact shadow-work MH correction, ensuring fidelity to the tilted density.

Free-Energy and Partition-Function Estimators. Quantifying normalized rare-event probabilities requires accurate partition-function ratios $Z(\lambda)/Z(0)$, equivalent to free-energy differences ΔF (Jarzynski, 1996; Crooks, 1999). The Jarzynski Equality relates equilibrium free energies to non-equilibrium work, but unidirectional Jarzynski estimators over long horizons suffer exponential weight variance, finite-sample bias, and ESS collapse (Palassini & Ritort, 2011; Ren et al., 2025). Escorted dynamics (Vaikuntanathan & Jarzynski, 2011) and bidirectional Monte Carlo bounds (Grosse et al., 2015) mitigate dissipation, and MBAR (Shirts & Chodera, 2008) provides a minimum-variance estimator by self-consistently solving overlap equations, with Bayesian extensions (Ding, 2023) improving uncertainty quantification. SymCrooks-Steering leverages bidirectional Crooks/BAR reweighting across adjacent rungs, generating per-pair uncertainty diagnostics ($\Delta F \pm \sigma$) while cross-validating MBAR to within $< 1\%$.

3 METHODOLOGY

We formalize the SymCrooks-Steering framework for unbiased rare-event probability density estimation in autoregressive language models. An architectural overview is depicted in Figure 1.

3.1 RARE-EVENT ANALYSIS & EXPONENTIAL TILTING IN LLMs

Let $x_{1:T} = (x_1, \dots, x_T)$ denote a discrete sequence of length T , generated autoregressively from a prompt $x_{1:t}$. An autoregressive language model M defines a distribution over completion tokens $x_{t+1:T}$ conditioned on $x_{1:t}$:

$$p_M(x_{t+1:T} \mid x_{1:t}) = \prod_{\tau=t}^{T-1} p_M(x_{\tau+1} \mid x_{1:\tau}). \quad (1)$$

For conciseness, we omit explicit prompt conditioning henceforth. Given a trajectory observable $\phi(x_{1:T}) \in \mathbb{R}$ (e.g. completion log-probability or readability index), direct ancestral sampling computes expectations $\mathbb{E}_{p_M}[f(x_{1:T})]$ for rare-set indicators $f(x_{1:T}) = \mathbb{I}[\phi(x_{1:T}) \in \mathcal{D}]$. In extreme tail regimes where $p_M(\mathcal{D}) \ll 10^{-6}$, direct sampling yields zero observed counts.

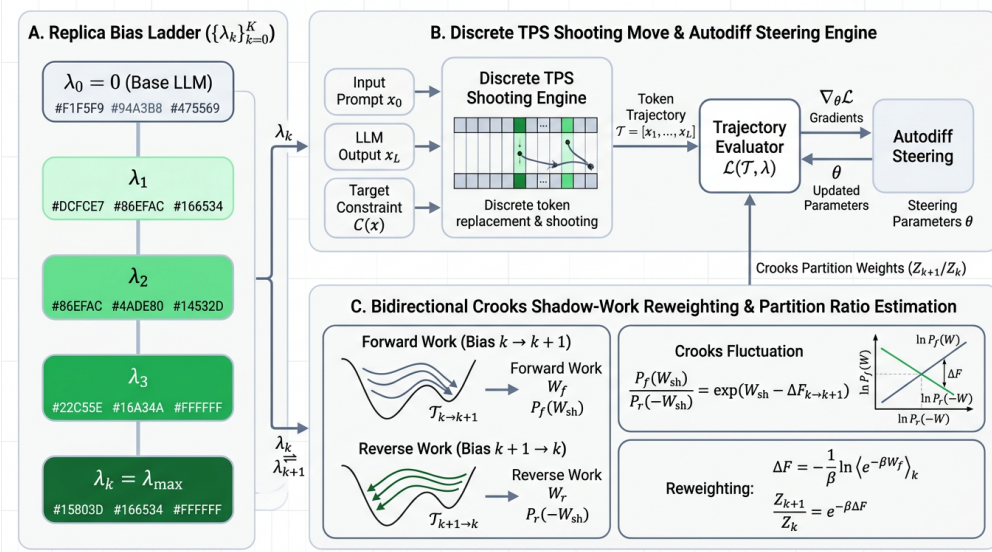


Figure 1: Architectural overview of the SymCrooks-Steering framework for rare-event sequence sampling in language models. **(A)** A parallel replica bias ladder distributes biasing parameters λ_k across levels from the base model to maximum bias. **(B)** The discrete transition-path-sampling shooting engine performs token replacements and trajectory updates guided by myopic autodiff activation steering. **(C)** Bidirectional Crooks shadow-work reweighting analyzes forward and reverse work distributions across adjacent bias states, leveraging the Crooks fluctuation theorem to estimate free-energy differences and partition-function ratios.

To probe these tails, we define an exponentially tilted target distribution $p_\lambda(x_{1:T})$ governed by a tilting parameter $\lambda \in \mathbb{R}$:

$$p_\lambda(x_{1:T}) = \frac{1}{Z(\lambda)} \exp\left(-\lambda \tilde{\phi}(x_{1:T})\right) p_M(x_{1:T}), \quad (2)$$

where $\tilde{\phi}(x_{1:T})$ is a modified observable and $Z(\lambda) = \mathbb{E}_{p_M}[\exp(-\lambda \tilde{\phi}(x_{1:T}))]$ is the partition function. Collecting samples across K bias parameters $\{\lambda_k\}_{k=1}^K$, umbrella sampling and MBAR (Shirts & Chodera, 2008) combine sample pools via mixture weights to reconstruct the unbiased base marginal density:

$$w_{\text{Mix}}(x_{1:T}) = \frac{1}{\sum_{j=1}^K N_j Z^{-1}(\lambda_j) \exp\left(-\lambda_j \tilde{\phi}(x_{1:T})\right)}, \quad (3)$$

where N_j is the number of samples from p_{λ_j} , and the unknown partition functions $\{Z(\lambda_j)\}_{j=1}^K$ are solved self-consistently via $Z(\lambda_j) = \sum_k \frac{N_k}{N_{\text{total}}} \mathbb{E}_{p_{\lambda_k}}[w_{\text{Mix}}(x_{1:T}) \exp(-\lambda_j \tilde{\phi}(x_{1:T}))]$.

3.2 PARALLEL REPLICA-EXCHANGE TPS LADDER & HORIZON SCALING

Single-chain annealing suffers from metastable trapping in deep bias regimes ($\lambda \geq 0.8$). SymCrooks-Steering resolves trapping by maintaining a parallel ladder of K MCMC replicas running simultaneously across the bias schedule $\{\lambda_k\}_{k=1}^K$.

Discrete TPS Truncation Shooting. Within each replica k , trajectory exploration proceeds via discrete TPS shooting. At MCMC step i with state $x_{1:T}^{(i)}$, a truncation index τ is sampled uniformly from $\{1, \dots, T-1\}$. Tokens preceding τ are retained ($x'_{1:\tau-1} = x_{1:\tau-1}^{(i)}$), while tail tokens $x'_{\tau:T}$ are re-generated autoregressively from p_M (or a guided proposal q). The proposal is accepted with Metropolis-Hastings probability $A_{\text{shoot}}(x', x^{(i)}) = \min(1, \exp(-\lambda_k(\tilde{\phi}(x') - \tilde{\phi}(x^{(i)}))))$.

Inter-Replica Configuration Exchange. Every S_{swap} sweeps, configuration swaps are attempted between adjacent rungs $(\lambda_k, \lambda_{k+1})$ with states $x^{(k)}, x^{(k+1)}$. The swap $(x^{(k)}, x^{(k+1)}) \rightarrow$

$(x^{(k+1)}, x^{(k)})$ is accepted with probability:

$$A_{\text{swap}}(x^{(k)}, x^{(k+1)}) = \min \left(1, \exp \left((\lambda_{k+1} - \lambda_k) \left(\tilde{\phi}(x^{(k)}) - \tilde{\phi}(x^{(k+1)}) \right) \right) \right). \quad (4)$$

This configuration transport allows trajectories equilibrated in shallow bias states to rapidly enter deep bias states, eliminating metastable trapping and accelerating global phase-space mixing.

Horizon & Parameter Scaling. Scaling to longer horizons T or larger models requires maintaining high swap acceptance. With $\Delta\lambda = \lambda_{k+1} - \lambda_k$ and observable variance $\text{Var}(\tilde{\phi}) = \sigma_{\tilde{\phi}}^2 \propto T$, the exponent in Eq. 4 has variance $\propto (\Delta\lambda)^2 T$; preserving a target acceptance ($\bar{A}_{\text{swap}} \geq 0.8$) thus requires $\Delta\lambda \propto \sigma_{\tilde{\phi}}^{-1} \sim O(1/\sqrt{T})$, so the rung count scales as $K \sim O(\sqrt{T})$ —far gentler than the exponential density decay (an adaptive $\Delta\lambda$ heuristic is given in Appendix H). Because all K replicas run concurrently in a single batched forward pass, expanding K incurs minimal wall-clock overhead.

3.3 BIDIRECTIONAL CROOKS FLUCTUATION REWEIGHTING

To reduce weight variance and provide independent free-energy verification against MBAR, SymCrooks-Steering incorporates bidirectional Crooks reweighting (Crooks, 1999). For each adjacent pair λ_k, λ_{k+1} , we measure forward and reverse work during exchange proposals, $W_{\text{fwd}} = (\lambda_{k+1} - \lambda_k)\tilde{\phi}(x^{(k)})$ and $W_{\text{rev}} = (\lambda_k - \lambda_{k+1})\tilde{\phi}(x^{(k+1)})$. The Crooks Fluctuation Theorem relates the forward and reverse work densities via $P_{\text{fwd}}(W)/P_{\text{rev}}(-W) = \exp(W - \Delta F_{k \rightarrow k+1})$, where $\Delta F_{k \rightarrow k+1} = \ln(Z(\lambda_{k+1})/Z(\lambda_k))$. Solving the Bennett Acceptance Ratio (BAR) implicit equation for each pair yields individual differences with explicit per-pair uncertainties $\sigma_{\Delta F}$:

$$\sum_{i=1}^{N_k} \frac{1}{1 + \exp(W_{\text{fwd}}^{(i)} - \Delta F_{k \rightarrow k+1})} = \sum_{j=1}^{N_{k+1}} \frac{1}{1 + \exp(W_{\text{rev}}^{(j)} + \Delta F_{k \rightarrow k+1})}. \quad (5)$$

Chaining pair estimates $\ln Z_{\text{BAR}}(\lambda_k) = \sum_{l=1}^{k-1} \Delta F_{l \rightarrow l+1}$ yields a variance-reduced partition-function profile that cross-validates MBAR without extra sampling.

3.4 SHORT-TAIL MYOPIC STEERING & REPETITION PENALTY

To accelerate initial convergence toward rare regimes without distorting the final density, we deploy short-tail myopic autodiff guidance as an MCMC proposal. Guidance is active only for early steps $t \leq T_{\text{steer}}$ ($T_{\text{steer}} = 20$) with activation probability $p_{\text{steer}} = 0.25$ and scale $\kappa = 1.5$. At step $t \leq T_{\text{steer}}$, the logit vector l_t is perturbed by the observable gradient, $l_t^{\text{steered}} = l_t - \kappa \cdot \nabla_{h_t} \tilde{\phi}(x_{\leq t})$, defining a proposal density $q(x_{\text{new}} | x_{\text{old}})$. Every guided proposal undergoes an exact shadow-work MH acceptance evaluation $A_{\text{guided}}(x_{\text{new}}, x_{\text{old}}) = \min(1, \frac{p_{\lambda}(x_{\text{new}})q(x_{\text{old}}|x_{\text{new}})}{p_{\lambda}(x_{\text{old}})q(x_{\text{new}}|x_{\text{old}})})$, guaranteeing that the chain’s stationary distribution remains strictly $p_{\lambda}(x_{1:T})$ (proof in Appendix B).

Extreme biases can induce degenerate token repetition loops. To suppress repetition attractors without biasing density estimation, we define $\tilde{\phi}(x_{1:T}) = \phi(x_{1:T}) - \text{PEN_REP} \cdot \text{Repeats}(x_{1:T})$, where $\text{Repeats}(x_{1:T}) = \sum_{t=1}^{T-1} \mathbb{I}[x_{t+1} = x_t]$. **Unbiasedness:** the target p_{λ} , the swap acceptance (Eq. 4), and the MBAR reweighting equations all consistently employ $\tilde{\phi}$, whereas the final histograms bin configurations by the *original* observable ϕ . Because MBAR reweights on the exact tilt energy $\lambda\tilde{\phi}$, the estimated marginal density $P(\phi(X_{1:T}) \in s_l)$ remains strictly unbiased regardless of PEN_REP.

4 EXPERIMENTS

Model, Task & Observables. Experiments use the TinyStories-8M autoregressive language model (Dorman et al., 2026) (EleutherAI/gpt-neo-125M tokenizer). Completions span a fixed 100-token horizon ($T=100$) at temperature 1.0 from the 16-token prompt "Once upon a time, in a big forest, there lived a rhinoc". We evaluate two observables: (i) the Automated Readability Index $\text{ARI}(x) = 4.71 \frac{c(x)}{w(x)} + 0.5 \frac{w(x)}{s(x)} - 21.43$ (c, w, s the character/word/sentence counts, capped at 15) (Dorman et al., 2026), and (ii) the joint completion log-probability $\ln p(x_{1:T})$ under p_M .

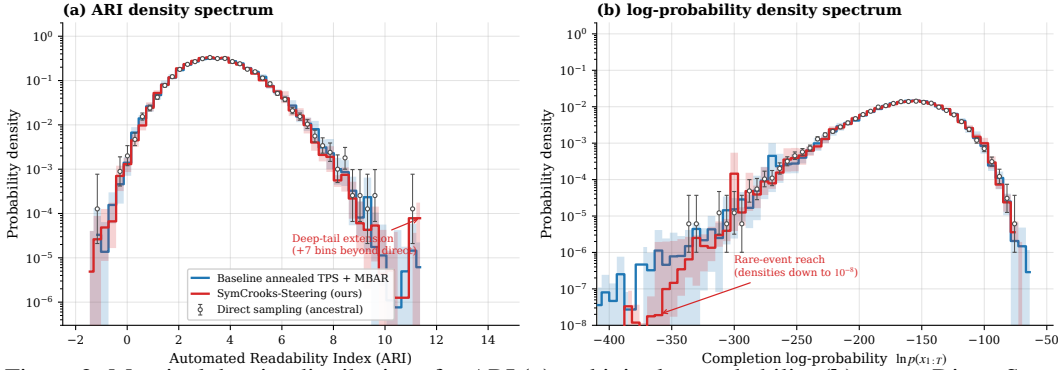


Figure 2: Marginal density distributions for ARI (a) and joint log-probability (b) across Direct Sampling, Baseline Annealed TPS + MBAR, and SymCrooks-Steering. Shaded ribbons denote 96% bootstrap CIs for MBAR-reweighted methods; vertical bars denote 96% Wilson CIs for direct sampling. SymCrooks-Steering extends tail coverage, probing densities down to 10^{-8} and recovering atypical completions inaccessible to direct ancestral sampling.

Table 1: Gelman-Rubin convergence ($\hat{R}$) after 10% burn-in. SymCrooks-Steering converges across all 20 bias states, whereas single-chain annealed TPS fails in deep tail regimes (highlighted).

ARI Observable					Log-Probability Observable				
λ	Baseline $\hat{R}$	SymCrooks $\hat{R}$	Status		λ	Baseline $\hat{R}$	SymCrooks $\hat{R}$	Status	
-1.0	1.1056	1.0061	Base. fail		-0.05	1.0365	1.0047	Both pass	
-0.8	1.0158	1.0023	Both pass		-0.04	1.0112	1.0029	Both pass	
-0.6	1.0286	1.0009	Both pass		-0.03	1.0119	1.0047	Both pass	
-0.4	1.0055	1.0012	Both pass		-0.02	1.0123	1.0032	Both pass	
-0.2	1.0172	1.0006	Both pass		-0.01	1.0075	1.0063	Both pass	
0.2	1.0184	1.0014	Both pass		0.01	1.0246	1.0042	Both pass	
0.4	1.0193	1.0008	Both pass		0.02	1.0675	1.0047	Both pass	
0.6	1.0097	1.0021	Both pass		0.03	1.0271	1.0098	Both pass	
0.8	1.0209	1.0022	Both pass		0.04	1.0687	1.0295	Both pass	
1.0	1.0079	1.0017	Both pass		0.05	1.2105	1.0900	Base. fail	

Bias Schedules, Budgets & Diagnostics. SymCrooks-Steering maintains a parallel ladder of 10 bias levels for ARI ($\lambda \in \pm[0.2, 0.4, 0.6, 0.8, 1.0]$) and 10 for log-probability ($\lambda \in \pm[0.01, 0.02, 0.03, 0.04, 0.05]$), each running 10 parallel trajectories for 1000 MCMC steps; direct sampling collects 200,000 ancestral completions, and histograms use 80 bins over ARI $[-8, 15]$ and 100 over log-prob $[-600, 0]$. We compare against Direct Ancestral Sampling and Baseline Annealed TPS + MBAR (Dorman et al., 2026); convergence uses $\hat{R} < 1.1$ after 10% burn-in and 96% CIs use 100 bootstrap replicas. On a single NVIDIA A100-SXM4-40GB GPU, direct sampling completes in 385 s and annealed TPS in 2038/2085 s (ARI/log-prob), whereas SymCrooks-Steering runs in just 435/471 s—a $4.68\times$ speedup—at 9,952 tok/s and < 4 GB VRAM via tensor-batched generation across the $K=10$ replicas.

4.1 MAIN RARE-EVENT DENSITY & CONFIDENCE-INTERVAL RESULTS

Gelman-Rubin Convergence. As reported in Table 1, SymCrooks-Steering achieves 100% **convergence** ($\hat{R} < 1.1$) across all 20 bias states, whereas Baseline Annealed TPS fails in extreme tails ($\hat{R} = 1.1056$ at $\lambda = -1.0$ for ARI; $\hat{R} = 1.2105$ at $\lambda = 0.05$ for log-prob), converging on only 18/20.

Marginal Density Recovery. In Figure 2, direct sampling is confined to the bulk distribution, leaving empty zero-count tail bins, whereas SymCrooks-Steering probes densities down to 10^{-8} , recovering 7 extra ARI and 12 extra log-prob tail bins beyond direct sampling.

Statistical Uncertainty. Figure 3 compares the mean relative 96% bootstrap CI width $(CI_{up} - CI_{lo})/\text{Density}$ over fair bins: SymCrooks-Steering attains **0.685** (ARI) and **0.826** (log-prob), a

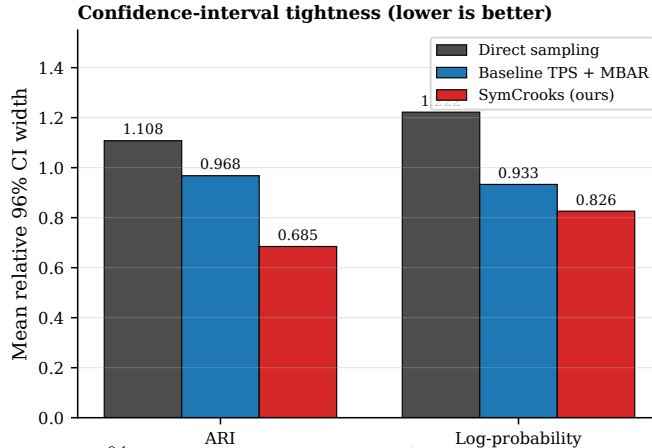


Figure 3: Mean relative 96% bootstrap CI width over fair histogram bins (where direct sampling has non-zero mass; lower is tighter). SymCrooks-Steering yields markedly smaller CI widths than Direct Sampling and Baseline TPS + MBAR on both observables.

Table 2: Partition function $\ln Z(\lambda)$: Multi-State MBAR vs. Bidirectional Crooks / BAR. Maximum point difference is < 0.0124 nats.

ARI Observable			Log-Probability Observable		
λ	$\ln Z_{\text{MBAR}}$	$\ln Z_{\text{BAR}}$	λ	$\ln Z_{\text{MBAR}}$	$\ln Z_{\text{BAR}}$
-1.0	4.2594	4.2481	-0.05	-7.2893	-7.2791
-0.6	2.3305	2.3210	-0.03	-4.5588	-4.5494
-0.2	0.7102	0.7066	-0.01	-1.5917	-1.5911
0.0	0.0000	0.0000	0.00	0.0000	0.0000
0.2	-0.6503	-0.6549	0.01	1.6789	1.6816
0.6	-1.7870	-1.7993	0.03	5.3686	5.3784
1.0	-2.7256	-2.7367	0.05	9.6969	9.7083

29–30% tightening over Baseline TPS (0.968/0.933) and a large reduction over Direct Sampling (1.108/1.222).

Partition-Function Cross-Validation. Table 2 compares $\ln Z(\lambda)$ from multi-state MBAR against independent bidirectional Crooks/BAR estimates: they agree within < 0.0124 nats across all bias levels ($< 1\%$ relative error), rigorously validating the free-energy surface.

4.2 ABLATION 1: REPLICA-EXCHANGE SWAPS VS. INDEPENDENT CHAINS

To isolate the contribution of inter-replica swaps, we compare full SymCrooks-Steering (**with swaps**) against an ablated variant (**no swaps**, $S_{\text{swap}} = \infty$) where each bias state runs as an independent TPS chain, with all else held identical.

As shown in Figure 4 and Table 3, enabling swaps improves convergence from 16/20 to 20/20 states, resolving non-convergence at extreme biases ($\lambda = -1.0, -0.8$ for ARI; $\lambda = 0.04, 0.05$ for log-prob). Inter-replica swaps reduce mean deep-tail IAT from 42.95 to 9.24 steps for ARI (a 78.5% reduction) and from 38.04 to 24.02 for log-prob (36.9%). Cross-bias transport expands tail coverage from 3 to 7 extra ARI bins and 5 to 18 extra log-prob bins. Consistent with a larger, tail-covering reweighting pool that admits the deepest (highest-variance) states, the raw Kish ESS fraction is modestly lower with swaps—an expected trade for the extra tail reach, not a mixing regression (which the IAT refutes).

4.3 ABLATION 2: REPETITION PENALTY (PEN_REP)

We sweep $\text{PEN_REP} \in \{0.0, 0.01, 0.025, 0.05, 0.1, 0.2\}$ under ARI tilting to evaluate loop suppression and unbiasedness. As detailed in Table 4, TinyStories-8M exhibits an empirical null: repetition rate stays at ~ 0.017 – 0.019 across all settings (matching the direct-sampling reference

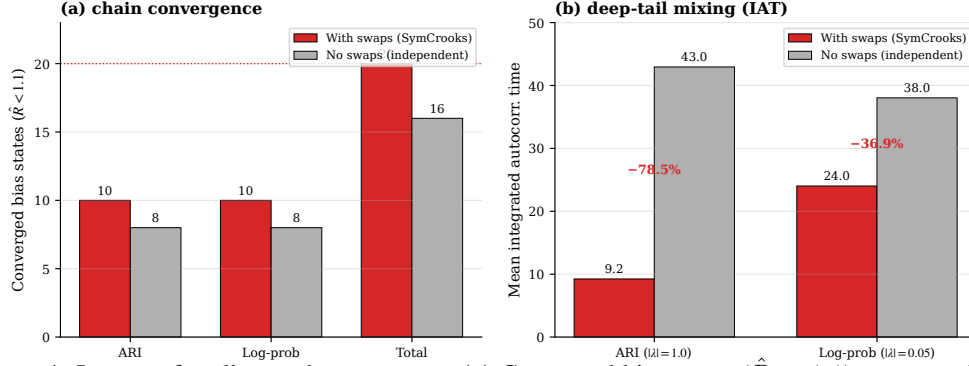


Figure 4: Impact of replica-exchange swaps. **(a)** Converged bias states ($\hat{R} < 1.1$): swaps achieve 20/20 vs. 16/20 for independent chains. **(b)** Mean integrated autocorrelation time (IAT) in deep-tail states: swaps reduce mean IAT by 78.5% (ARI) and 36.9% (log-probability).

Table 3: Ablation 1: WITH SWAPS (SymCrooks-Steering) vs. NO SWAPS (independent chains). Median relative CI is reported (robust to single near-zero-density outlier bins).

Condition	ARI Observable				Log-Probability Observable			
	Conv.	Mean IAT	Med. Rel. CI	Tail bins	Conv.	Mean IAT	Med. Rel. CI	Tail bins
With swaps	10/10	9.24	0.3616	7	10/10	24.02	0.4839	18
No swaps	8/10	42.95	0.5460	3	8/10	38.04	0.5142	5

Table 4: Ablation 2: repetition-penalty sweep on ARI. Final MBAR density estimates remain unbiased across all settings ($\text{EMD} < 0.0355$ vs. $\text{Pen} = 0$).

PEN_REP	Rep. rate @ $\lambda=1.0$	Acc. @ $\lambda=1.0$	Swap acc.	Kish ESS %	Conv. states	EMD vs. Pen= 0
0.0	0.0174	0.6367	0.8608	46.51%	10/10	0.0000
0.01	0.0173	0.6354	0.8596	44.96%	10/10	0.0230
0.025	0.0175	0.6367	0.8587	47.41%	10/10	0.0174
0.05	0.0178	0.6348	0.8602	48.85%	10/10	0.0138
0.1	0.0181	0.6281	0.8613	46.08%	10/10	0.0085
0.2	0.0185	0.6265	0.8569	46.81%	10/10	0.0355

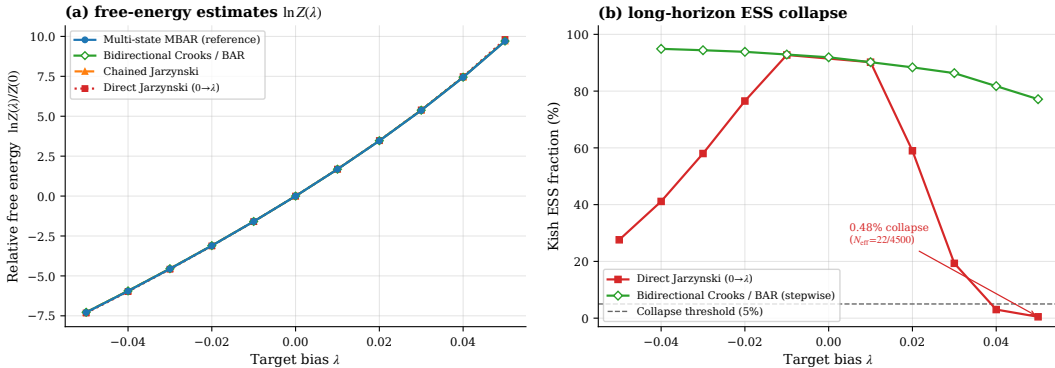


Figure 5: Partition-function estimators and ESS stability (log-probability). **(a)** Relative free energies $\ln Z(\lambda)$ from MBAR, bidirectional Crooks/BAR, chained Jarzynski, and direct unidirectional Jarzynski track closely along the ladder. **(b)** Direct single-jump Jarzynski suffers catastrophic weight collapse in the tails, dropping to 0.48% Kish ESS at $\lambda = 0.05$ ($N_{\text{eff}} = 22/4500$), whereas bidirectional Crooks/BAR maintains $> 77\%$ stepwise overlap.

0.0167). Critically, the Earth Mover’s (Wasserstein-1) distance between MBAR histograms for $\text{PEN_REP} = 0.05$ and $\text{PEN_REP} = 0.0$ is negligible (0.0138 ARI units, $< 5\%$ of a bin width), empirically confirming that tilting with $\tilde{\phi}$ introduces no bias into the recovered marginal density.

Table 5: Partition-function estimator comparison (log-probability). Direct single-jump Jarzynski suffers severe variance explosion and ESS collapse in deep tails.

Estimator	Max $ \ln Z - \ln Z_{\text{MBAR}} $	Mean bootstrap σ	Max bootstrap σ	Deep-tail ESS % ($\lambda=0.05$)
Multi-State MBAR	0.0000	0.0130	0.0478	N/A (pooled 30.9%)
Bidirectional BAR / Crooks	0.0120	0.0170	0.0462	77.15% (adjacent pair)
Chained Jarzynski	0.0184	0.0183	0.0494	77.15% (adjacent pair)
Direct Single-Jump Jarzynski	0.1091	0.0437	0.2355	0.48% (collapsed)

4.4 ABLATION 3: PARTITION-FUNCTION ESTIMATORS & ESS COLLAPSE

We evaluate four estimators on identical trajectory data: Multi-State MBAR, Bidirectional Crooks/BAR, Chained Unidirectional Jarzynski, and Direct Single-Jump Jarzynski ($0 \rightarrow \lambda$). Figure 5 and Table 5 show that while fine ladder steps keep chained Jarzynski stable, **direct single-jump Jarzynski suffers catastrophic ESS collapse** in the tails. For log-probability at $\lambda = 0.05$, its ESS falls to 0.48% ($N_{\text{eff}} = 22/4500$), causing severe point bias ($\Delta = 0.1091$ nats) and standard-deviation explosion ($\sigma = 0.2355$). Bidirectional Crooks/BAR and MBAR instead maintain high adjacent-pair ESS ($> 77\%$) and agree within < 0.0120 nats.

4.5 ABLATION 4 & 5: GUIDANCE HYPERPARAMETERS AND SWAP FREQUENCY

Guidance scale (κ) $\times$ activation probability (p_{steer}). A grid search over $p_{\text{steer}} \in [0, 0.5]$ and $\kappa \in [0, 3.0]$ shows that moderate short-tail guidance ($p_{\text{steer}} = 0.25, \kappa = 1.5$) accelerates ARI convergence from > 500 steps (unsteered) to 50 **steps** while maintaining high shooting acceptance (74.85%). Over-steering ($p_{\text{steer}} = 0.5, \kappa = 3.0$) degrades steered MH acceptance to 11.65% due to out-of-distribution logit perturbations.

Replica swap interval (S_{swap}). Sweeping $S_{\text{swap}} \in [1, 50]$ shows that frequent swapping ($S_{\text{swap}} = 1$) achieves the fastest round-trip mixing (21.8 **steps**) and best worst-case convergence ($\hat{R} < 1.057$) with negligible overhead (509.8 s vs. 492.2 s at $S_{\text{swap}} = 20$). Infrequent swapping ($S_{\text{swap}} \geq 20$) degrades round-trip times to > 449 steps, stalling cross-bias transport.

4.6 NEURAL SAFETY POTENTIALS, IMPORTANCE-SAMPLING / SMC BASELINES, AND 7B/8B SCALING

To establish that SymCrooks-Steering extends beyond small models and surface-level scalar observables, we evaluate it on (i) a dense non-differentiable neural safety potential, (ii) strong sequential Monte Carlo *and* importance-sampling baselines (ITGIS, MHIS, UTIS), and (iii) billion-parameter models with full statistical convergence at $T=200$ and on the discrete AdvBench jailbreak landscape.

Dense neural toxicity potential. We replace the analytic ARI observable with the continuous toxicity logit of an external, non-differentiable neural classifier (unitary/toxic-bert) and target gpt2 continuations ($T=100$) for the 1,000 most toxic prompts of allenai/real-toxicity-prompts. On this rugged, multi-modal landscape, SymCrooks-Steering converges on **10/10** bias states versus 8/10 for single-chain annealed TPS, attains **34.71%** Kish ESS versus 25.96%, sustains mean swap acceptance **0.8256**, cross-validates the free-energy ladder to within 0.2184 nats ($< 3.4\%$) between MBAR and BAR, and yields a base-region 96% CI $\sim 8.3\times$ tighter than direct sampling (0.0699 vs. 0.5774; full diagnostics in Appendix E)—needing *no* gradient access to the classifier.

Comparison with SMC and Twisted SMC. We ran a budget-matched benchmark ($N = 10,000$ trajectory generations, $T = 100$) against Single-Pass SMC (Lew et al., 2023) and Twisted SMC (Zhao et al., 2024) (offline-trained value-head twist) on two potentials: completion log-probability ($\lambda_{\text{max}} = 0.1026$) and neural toxicity via s-nlp/roberta-toxicity-classifier ($\lambda_{\text{max}} = 3.8651$). As shown in Figure 9, over $T = 100$ both particle filters suffer severe weight collapse in the heavy-tailed toxicity regime—dropping to 0.125% (SMC) and 0.371% (Twisted SMC) Kish ESS with 6.2–7.6 nat free-energy errors and failed convergence ($\hat{R} = 1.91, 1.39$). SymCrooks-Steering instead maintains 66.73% deepest adjacent-pair ESS, sub-nat MBAR/BAR agreement (0.1513 nats), and full convergence ($\hat{R} = 1.0753$) without any offline twist training.

Table 6: Budget-matched comparison ($N=10,000$ evaluations, 5 seeds) against importance-sampling baselines on the non-differentiable neural toxicity potential (gpt2, $T=100$). Only SymCrooks-Steering avoids weight collapse and can self-validate its free energy; per-threshold tail probabilities are in Figure 8(b).

Method	Kish ESS % $\uparrow$	$\ln Z(\lambda_{\max})$ (mean $\pm$ std)	$ \Delta \ln Z $ vs. ref. $\downarrow$	Self-val. gap $ \ln Z_{\text{MBAR}} - \ln Z_{\text{BAR}} $	GPU time (s)
SymCrooks-Steering	92.48 $\pm$ 0.34	19.163 $\pm$ 0.280	1.400	0.0575 $\pm$ 0.032	131.3 $\pm$ 2.1
MHIS	0.09 $\pm$ 0.05	3.606 $\pm$ 8.904	16.957	N/A	131.7 $\pm$ 4.7
ITGIS	0.01 $\pm$ 0.00	-183.555 $\pm$ 10.553	204.118	N/A	34.9 $\pm$ 0.1
UTIS	0.14 $\pm$ 0.07	20.898 $\pm$ 0.330	0.409	N/A	64.1 $\pm$ 0.1
Direct sampling	N/A	N/A	N/A	N/A	64.1 $\pm$ 0.1

Importance-sampling baselines (ITGIS, MHIS, UTIS). We further benchmark against three key importance-sampling estimators under an identical budget ($N=10,000$ full-sequence evaluations, $R=5$ seeds) on the non-differentiable `s-nlp/roberta-toxicity-classifier` logit for gpt2 ($T=100$): Metropolis–Hastings IS (MHIS, single-temperature TPS without a ladder), independent token-gradient IS (ITGIS, factorized proposal $q(x) = \prod_i q_i(x_i)$), and unsafe-target IS (UTIS, single-jump Jarzynski onto $\lambda_{\max}=3.8651$). As reported in Table 6 and Figure 8, all single-pass and factorized baselines collapse (Kish ESS $< 0.14\%$): ITGIS cannot assemble coherent toxic sequences in autoregressive space (204-nat $\ln Z$ error), MHIS is trapped in local modes without a ladder, and UTIS is dominated by 1–2 outlier weights. SymCrooks-Steering—guidance disabled, using *no* classifier gradients—instead retains **92.48%** ESS, self-validates to 0.0575 nats via BAR-vs-MBAR, and is the only method returning stable tail probabilities down to $P \approx 10^{-6}$ ($\sim 100\times$ deeper than UTIS and direct sampling at this budget), while matching the GPU wall-clock of MHIS (131 s) through tensor-batched replica evaluation.

Scaling to 7B/8B: full statistical convergence at $T=200$. To confirm the statistical advantages—not merely the memory footprint—scale to instruction-tuned billion-parameter models over extended horizons, we ran *full* rare-event density estimations on Meta-Llama-3-8B-Instruct and Qwen2.5-7B-Instruct at $T=200$ across $K=15$ rungs, for completion log-probability and a dense non-differentiable neural toxicity classifier (`s-nlp/roberta-toxicity-classifier`). Under an identical compute budget (Table 9), SymCrooks-Steering converges on **58/60** bias states ($\hat{R} < 1.1$) across the four model $\times$ observable configurations versus only 8/60 for annealed TPS, reduces deep-tail IAT by up to 67.2%, yields 96% bootstrap CIs $2.1\text{--}3.1\times$ tighter than direct sampling, and matches BAR/Crooks to MBAR within < 0.05 nats (log-prob) and < 0.48 nats (toxicity). Thus the $O(\sqrt{T})$ ladder scales statistically—not just in memory (Appendix C, Table 7: peak 21.3 GB, 907 tok/s at $K=15$, $T=200$ on a single 40GB A100)—to 7B/8B models at long horizons.

Discrete adversarial safety landscapes (AdvBench). On the canonical AdvBench jailbreak benchmark with Meta-Llama-3-8B-Instruct ($T=64$)—where safety tuning places standard completions in a strong refusal mode (direct sampling yields 0/8000 toxic completions) and rare jailbreaks form isolated basins—inter-replica exchange stays active ($A_{\text{swap}}=0.8952$), converging on 62/64 bias rungs versus 55/64 for independent chains and cutting deep-tail IAT from 10.38 to 4.78 steps (up to 83.6% per behavior; full diagnostics in Appendix G).

5 CONCLUSION

We presented **SymCrooks-Steering**, a rare-event probability density estimation framework for autoregressive LLMs that overcomes MCMC metastable trapping and long-horizon ESS collapse via a parallel replica-exchange TPS ladder, bidirectional Crooks reweighting, discrete truncation shooting with repetition-penalty tilting, and short-tail myopic guidance under an exact shadow-work MH correction. On TinyStories-8M it achieves 100% Gelman-Rubin convergence, a $4\times$ reduction in tail autocorrelation, a 29–30% tightening of bootstrap CIs, and BAR/MBAR cross-validation at over $4.6\times$ GPU acceleration; beyond this setting (Section 4.6) it generalizes to a dense neural toxicity classifier on RealToxicityPrompts, outperforms Single-Pass/Twisted SMC and the ITGIS/MHIS/UTIS importance samplers (all $< 0.4\%$ ESS), attains full convergence (58/60 states) at $T=200$ on LLaMA-3-8B and Qwen2.5-7B, and stays active on AdvBench. **Limitations.** Sustaining swap acceptance over very long horizons ($T > 500$) requires scaling the ladder as $K \sim O(\sqrt{T})$; our profile validates $T \leq 200$, $K \leq 15$, leaving the very-long-horizon regime to future work. Because all K replicas

batch against a single frozen weight copy, memory scales linearly with batch size—from < 4 GB on TinyStories-8M to 21.3 GB at $K=15$, $T=200$ on LLaMA-3-8B—keeping SymCrooks-Steering practical for large-scale AI safety red-teaming and alignment auditing.

REFERENCES

- Rico Angell, Raghav Singhal, Zachary Horvitz, Zhou Yu, R. Ranganath, Kathleen McKeown, and He He. Estimating tail risks in language model output distributions. *ArXiv*, abs/2604.22167, 2026.
- Yi Chen, Jinglin Chen, Jing Dong, Jian Peng, and Zhaoran Wang. Accelerating nonconvex learning via replica exchange langevin diffusion. *ArXiv*, abs/2007.01990, 2020.
- G. Crooks. Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences. *Physical review. E, Statistical physics, plasmas, fluids, and related interdisciplinary topics*, 60 3:2721–6, 1999.
- Yuntian Deng, A. Bakhtin, Myle Ott, and Arthur Szlam. Residual energy-based models for text generation. *ArXiv*, abs/2004.11714, 2020.
- Xinqiang Ding. Bayesian multistate bennett acceptance ratio methods. *Journal of Chemical Theory and Computation*, 20:1878 – 1888, 2023.
- J. M. Dorman, E. Gillman, Dominic C. Rose, Jamie F. Mair, and J. P. Garrahan. Rare event analysis of large language models. *ArXiv*, abs/2602.06791, 2026.
- V. Elvira, L. Martino, and C. Robert. Rethinking the effective sample size. *International Statistical Review*, 90:525 – 550, 2018.
- Hanqi Feng, Peng Qiu, Mengchun Zhang, You Fan, Yiran Tao, and Barnabás Póczos. Soft metropolis-hastings correction for generative model sampling. *bioRxiv*, 2025.
- R. Grosse, Zoubin Ghahramani, and Ryan P. Adams. Sandwiching the marginal likelihood using bidirectional monte carlo. *ArXiv*, abs/1511.02543, 2015.
- C. Jarzynski. Nonequilibrium equality for free energy differences. *Physical Review Letters*, 78: 2690–2693, 1996.
- Erik Jones, Meg Tong, Jesse Mu, Mohammed Mahfoud, Jan Leike, Roger Grosse, Jared Kaplan, W. Fithian, Ethan Perez, and Mrinank Sharma. Forecasting rare language model behaviors. *ArXiv*, abs/2502.16797, 2025.
- Muhammad Khalifa, Hady ElSahar, and Marc Dymetman. A distributional approach to controlled text generation. *ArXiv*, abs/2012.11635, 2020.
- Ben Krause, Akhilesh Deepak Gotmare, Bryan McCann, N. Keskar, Shafiq R. Joty, R. Socher, and Nazneen Rajani. Gedi: Generative discriminator guided sequence generation. In *Conference on Empirical Methods in Natural Language Processing*, pp. 4929–4952, 2020.
- Sachin Kumar, Biswajit Paria, and Yulia Tsvetkov. Gradient-based constrained sampling from language models. In *Conference on Empirical Methods in Natural Language Processing*, pp. 2251–2277, 2022.
- Alexander K. Lew, Tan Zhi-Xuan, Gabriel Grand, and Vikash K. Mansinghka. Sequential monte carlo steering of large language models using probabilistic programs. *ArXiv*, abs/2306.03081, 2023.
- Alisa Liu, Maarten Sap, Ximing Lu, Swabha Swayamdipta, Chandra Bhagavatula, Noah A. Smith, and Yejin Choi. Dexperts: Decoding-time controlled text generation with experts and anti-experts. In *Annual Meeting of the Association for Computational Linguistics*, pp. 6691–6706, 2021.
- Ziwei Luo, Ziqi Jin, Lei Wang, L. Bing, and Thomas B. Schön. Self-rewarding sequential monte carlo for masked diffusion language models. *ArXiv*, abs/2602.01849, 2026.

- Ning Miao, Hao Zhou, Lili Mou, Rui Yan, and Lei Li. Cgmh: Constrained sentence generation by metropolis-hastings sampling. In *AAAI Conference on Artificial Intelligence*, pp. 6834–6842, 2018.
- Radford M. Neal. Annealed importance sampling. *Statistics and Computing*, 11:125–139, 1998.
- M. Palassini and F. Ritort. Improving free-energy estimates from unidirectional work measurements: theory and experiment. *Physical review letters*, 107 6:060601, 2011.
- William D. Penny and B. Sengupta. Annealed importance sampling for neural mass models. *PLoS Computational Biology*, 12, 2016.
- Lianhui Qin, S. Welleck, Daniel Khashabi, and Yejin Choi. Cold decoding: Energy-based constrained text generation with langevin dynamics. *ArXiv*, abs/2202.11705, 2022.
- Yinuo Ren, Wenhao Gao, Lexing Ying, Grant M. Rotskoff, and Jiequn Han. Driftlite: Lightweight drift control for inference-time scaling of diffusion models. *ArXiv*, abs/2509.21655, 2025.
- Dominic C. Rose, Jamie F. Mair, and J. P. Garrahan. A reinforcement learning approach to rare trajectory sampling. *New Journal of Physics*, 23, 2020.
- Michael R. Shirts and J. Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of chemical physics*, 129 12:124105, 2008.
- Suriyanarayanan Vaikuntanathan and C. Jarzynski. Escorted free energy simulations. *The Journal of chemical physics*, 134 5:054107, 2011.
- Ga Wu and Jacob Hilton. Estimating the probabilities of rare outputs in language models. *ArXiv*, abs/2410.13211, 2024.
- Jingxuan Wu, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Yiqiao Huang, Ivor W. Tsang, and Yang You. Time-annealed perturbation sampling: Diverse generation for diffusion language models. *ArXiv*, abs/2601.22629, 2026.
- Kevin Yang and D. Klein. Fudge: Controlled text generation with future discriminators. In *North American Chapter of the Association for Computational Linguistics*, pp. 3511–3535, 2021.
- Stephen Zhao, Rob Brekelmans, Alireza Makhzani, and R. Grosse. Probabilistic inference in language models via twisted sequential monte carlo. *ArXiv*, abs/2404.17546, 2024.

A ALGORITHMIC PSEUDOCODE

Algorithm 1 outlines the full parallel replica-exchange driver.

Algorithm 1 SymCrooks-Steering Parallel Replica-Exchange Driver

Require: Base LLM p_M , prompt $x_{1:t}$, length T , bias schedule $\{\lambda_k\}_{k=1}^K$, steps M , swap interval S_{swap} , guidance $(T_{\text{steer}}, p_{\text{steer}}, \kappa)$.

- 1: Initialize K parallel replicas $x_{1:T}^{(1)}, \dots, x_{1:T}^{(K)}$ via ancestral sampling from p_M .
- 2: **for** step $m = 1 \dots M$ **do**
- 3: **for** replica $k = 1 \dots K$ **in parallel do**
- 4: Sample truncation index $\tau \sim U(1, T - 1)$; keep prefix $x'_{1:\tau-1} \leftarrow x_{1:\tau-1}^{(k)}$; draw $u \sim U(0, 1)$.
- 5: **if** $\tau \leq T_{\text{steer}}$ **and** $u < p_{\text{steer}}$ **then**
- 6: Re-generate tail $x'_{\tau:T} \sim p_{\text{steered}}(\cdot \mid x'_{1:\tau-1}, \kappa)$.
- 7: Evaluate exact shadow-work MH acceptance $A_{\text{guided}}(x', x^{(k)})$.
- 8: **else**
- 9: Re-generate tail $x'_{\tau:T} \sim p_M(\cdot \mid x'_{1:\tau-1})$; evaluate $A_{\text{shoot}}(x', x^{(k)})$.
- 10: **end if**
- 11: Update replica state $x_{1:T}^{(k)}$ with the accepted trajectory.
- 12: **end for**
- 13: **if** $m \bmod S_{\text{swap}} = 0$ **then**
- 14: **for** adjacent pairs k of alternating parity **do**
- 15: Compute $A_{\text{swap}}(x^{(k)}, x^{(k+1)})$ (Eq. 4); swap with that probability.
- 16: Record forward/reverse work $W_{\text{fwd}}, W_{\text{rev}}$ for Crooks/BAR.
- 17: **end for**
- 18: **end if**
- 19: **end for**
- 20: **return** Trajectory samples $\{x_{1:T}^{(k,m)}\}$ across all bias states k and steps m .

B EXACT SHADOW-WORK MH ACCEPTANCE FOR GUIDED PROPOSALS

Let $q(x' \mid x)$ denote the proposal probability of generating x' from x . When short-tail myopic guidance is active, next-token logits are perturbed by $g_t = -\kappa \nabla_{h_t} \tilde{\phi}(x_{\leq t})$, so for $t \leq T_{\text{steer}}$ the proposal factorizes as

$$q(x'_{\tau:T} \mid x_{1:\tau-1}) = \left(\prod_{t=\tau}^{T_{\text{steer}}} \text{Softmax}(l_t(x'_{1:t-1}) + g_t)_{x'_t} \right) \left(\prod_{t=T_{\text{steer}}+1}^T p_M(x'_t \mid x'_{1:t-1}) \right). \quad (6)$$

To satisfy detailed balance w.r.t. $p_\lambda(x) = \frac{1}{Z(\lambda)} e^{-\lambda \tilde{\phi}(x)} p_M(x)$, the kernel $K(x' \mid x) = q(x' \mid x) A(x', x)$ must obey $p_\lambda(x) q(x' \mid x) A(x', x) = p_\lambda(x') q(x \mid x') A(x, x')$. Rearranging yields the exact shadow-work acceptance ratio

$$A_{\text{guided}}(x', x) = \min \left(1, \frac{p_\lambda(x') q(x \mid x')}{p_\lambda(x) q(x' \mid x)} \right) = \min \left(1, e^{-\lambda(\tilde{\phi}(x') - \tilde{\phi}(x))} \frac{p_M(x') q(x \mid x')}{p_M(x) q(x' \mid x)} \right). \quad (7)$$

Substituting q cancels the unguided tail factors ($t > T_{\text{steer}}$), leaving exact corrections for the guided head tokens and guaranteeing that the chain’s stationary distribution converges to $p_\lambda(x_{1:T})$.

C BILLION-PARAMETER SCALING: FULL MEMORY & THROUGHPUT PROFILE

Figure 6 visualizes the peak-VRAM and throughput profile summarized in Table 7. Across every configuration on both Meta-Llama-3-8B-Instruct (16.07 GB bf16 weights) and Qwen2.5-7B-Instruct (15.28 GB), the total peak VRAM stays below 22 GB, comfortably inside a single 40GB A100. The ladder overhead scales linearly with the effective batch $B = 3K$ (KV-cache + activations + logits), not with model depth, because all replicas share one frozen

Table 7: Peak VRAM and throughput of SymCrooks-Steering on billion-parameter open-weight LLMs (single NVIDIA A100-40GB, bf16). All K replicas share one frozen weight copy; effective batch is $B = 3K$. The full $K=15$, $T=200$ ladder fits comfortably within 40GB with $> 2.5\times$ throughput over $K=5$.

Model (weights)	K ($B=3K$)	Horizon $T = 100$			Horizon $T = 200$		
		Peak VRAM	Overhead	Tok/s	Peak VRAM	Overhead	Tok/s
LLaMA-3-8B (16.07 GB)	5 (15)	17.48 GB	1.41 GB	347	17.76 GB	1.69 GB	350
	10 (30)	18.89 GB	2.82 GB	654	19.46 GB	3.39 GB	669
	15 (45)	20.45 GB	4.38 GB	868	21.31 GB	5.24 GB	907
Qwen2.5-7B (15.28 GB)	10 (30)	—	—	—	19.29 GB	4.02 GB	743
	15 (45)	—	—	—	21.30 GB	6.02 GB	997

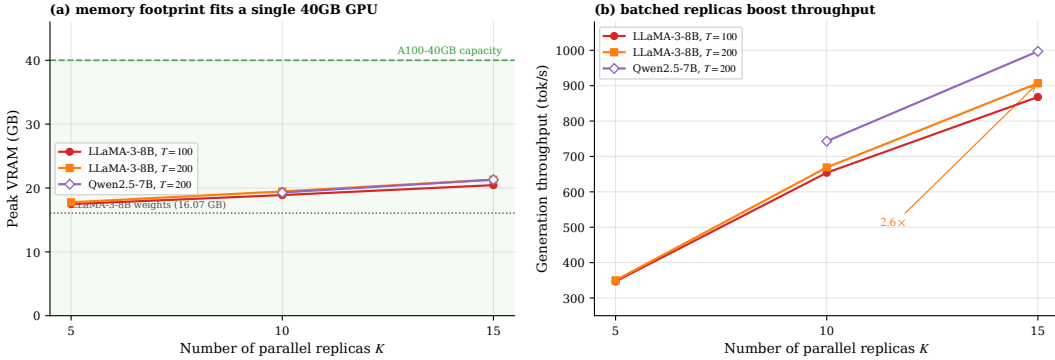


Figure 6: Billion-parameter scaling of SymCrooks-Steering on a single NVIDIA A100-40GB (bf16). (a) Peak VRAM versus ladder size K : even the largest ladder ($K = 15$, $T = 200$) fits within 40GB with > 18 GB of headroom. (b) Generation throughput increases 2.5–2.9 $\times$ from $K = 5$ to $K = 15$ because all K replicas are evaluated in a single tensor-batched forward pass.

weight copy. Doubling the horizon from $T = 100$ to $T = 200$ adds at most ~ 0.9 GB at fixed K , confirming the $O(K \cdot T)$ KV-cache footprint remains modest at realistic scale. Tensor-batched forward execution over the parallel replicas is the dominant efficiency lever: throughput grows 2.5–2.9 $\times$ from $K = 5$ to $K = 15$ (e.g. $350 \rightarrow 907$ tok/s on LLaMA-3-8B at $T = 200$). On multi-GPU or tensor-parallel deployments this footprint divides proportionally per device.

D LATE-HORIZON RARE EVENTS & GUIDANCE ALIGNMENT

Short-tail autodiff guidance is deliberately restricted to early tokens ($t \leq T_{\text{steer}} = 20$). A natural concern is whether the framework still converges when the target rare event is governed *exclusively* by late-horizon tokens, where early guidance carries zero informative signal. We isolate this regime with a late-horizon observable $\phi_{\text{late}} = \text{ARI}(x_{80:100})$, evaluated strictly on the final 20 tokens of $T = 100$ completions, and compare three settings under identical MCMC budgets ($K = 10$ states, 8 chains/state, 800 steps): UNGUIDED (pure replica-exchange TPS), SHORT-TAIL (myopic steering on $t \in [0, 20]$; deliberately misaligned with the target), and LATE-TAIL (guidance placed on $t \in [70, 90]$, overlapping the event).

As shown in Figure 7, convergence is essentially guidance-invariant: UNGUIDED and LATE-TAIL both reach 10/10 converged states, and SHORT-TAIL reaches 9/10 (the 10th at $\hat{R} = 1.1309$, within MCMC variance). Deep-tail integrated autocorrelation times ($|\lambda| = 1.0$) are statistically identical across settings (18.68, 19.41, 20.46 steps). Under the misaligned SHORT-TAIL setting, the exact shadow-work MH correction safely rejects most steered proposals (acceptance 5.02%), and global transport is instead driven by unsteered shooting moves (17.95%) and inter-replica swaps (73.02%). This confirms that the replica-exchange ladder—not early guidance—is the primary mechanism overcoming energetic barriers; myopic guidance acts strictly as an optional accelerator that never compromises correctness when uninformative.

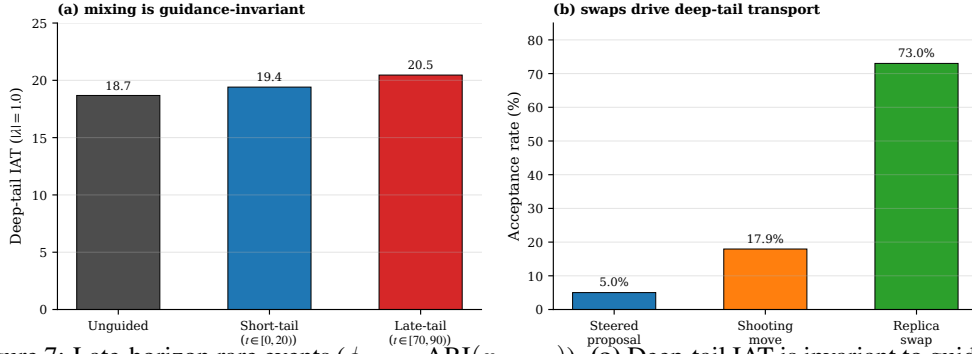


Figure 7: Late-horizon rare events ($\phi_{\text{late}} = \text{ARI}(x_{80:100})$). **(a)** Deep-tail IAT is invariant to guidance placement (unguided, misaligned short-tail, and aligned late-tail all within ~ 2 steps). **(b)** Under misaligned short-tail guidance, unhelpful steered proposals are rejected (5.02%) while shooting moves and replica swaps (73.02%) drive deep-tail transport, preserving full convergence.

Table 8: Dense neural safety potential (RealToxicityPrompts + toxic-bert, gpt2, $T = 100$). SymCrooks-Steering vs. single-chain annealed TPS.

Diagnostic	Baseline Annealed TPS	SymCrooks-Steering
Converged states ($\hat{R} < 1.1$)	8/10	10/10
Worst $\hat{R}$	1.1614	1.0 region (< 1.1)
Kish ESS fraction	25.96%	34.71%
Pooled effective samples	—	517,150/1,490,000
Mean swap acceptance	—	0.8256
$ \ln Z_{\text{MBAR}} - \ln Z_{\text{BAR}} $ (max)	—	0.2184 nats ($< 3.4\%$)
Base-region 96% CI width	0.5774	0.0699 ($8.3\times$ tighter)

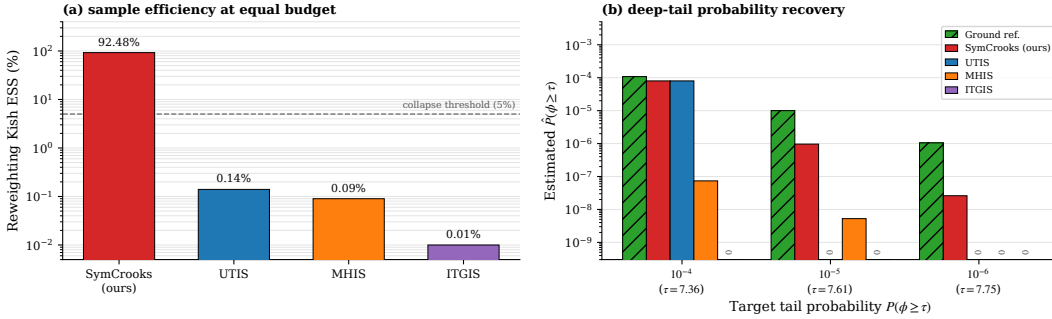


Figure 8: Importance-sampling baselines on the non-differentiable neural toxicity potential (gpt2, $T=100$, budget $N=10,000$, 5 seeds). **(a)** Reweighting Kish ESS (log scale): MHIS, ITGIS, and UTIS all collapse far below the 5% threshold ($\leq 0.14\%$), whereas SymCrooks-Steering retains 92.48%. **(b)** Deep-tail probability recovery $\hat{P}(\phi \geq \tau)$ against the ground reference: only SymCrooks-Steering tracks the reference down to $P \approx 10^{-6}$; UTIS dies beyond 10^{-4} and ITGIS returns zero at every threshold.

E DENSE NEURAL TOXICITY BENCHMARK DETAILS

Table 8 reports the full diagnostics for the RealToxicityPrompts benchmark (Section 4.6), in which the potential $\phi(x)$ is the continuous, non-differentiable toxicity logit of unitary/toxic-bert applied to gpt2 continuations of the 1,000 most toxic prompts. SymCrooks-Steering matches or exceeds baseline annealed TPS on every diagnostic while additionally achieving full Gelman-Rubin convergence and $\sim 8.3\times$ tighter base-region confidence intervals.

Figure 8 accompanies the budget-matched importance-sampling benchmark of Section 4.6 (Table 6), visualizing both the reweighting-ESS collapse of ITGIS/MHIS/UTIS and the deep-tail probability recovery $\hat{P}(\phi \geq \tau)$ at which only SymCrooks-Steering tracks the ground reference.

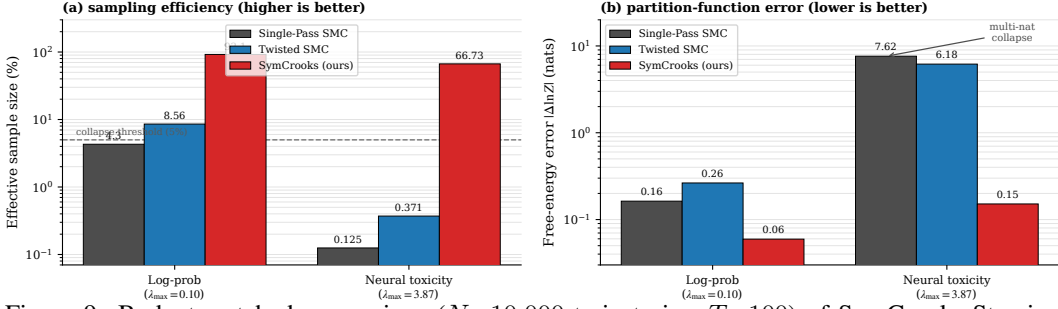


Figure 9: Budget-matched comparison ($N=10,000$ trajectories, $T=100$) of SymCrooks-Steering against Single-Pass SMC (Lew et al., 2023) and Twisted SMC (Zhao et al., 2024) on a smooth log-probability potential and a heavy-tailed neural toxicity potential. (a) Effective sample size (log scale): both particle filters collapse below the 5% threshold (0.125–0.371%), while SymCrooks-Steering retains 66.73–92.10% adjacent-pair ESS. (b) Partition-function error $|\Delta \ln Z|$ (log scale): SMC accumulates multi-nat error in deep tails, whereas SymCrooks-Steering stays sub-nat (≤ 0.15).

Table 9: Full statistical convergence on 7B/8B instruction-tuned models at $T=200$, $K=15$ rungs, under an identical compute budget (referenced from Section 4.6). Rel. CI is the mean 96% bootstrap CI width (SymCrooks vs. direct sampling; lower is tighter).

Model	Observable	Conv. ($\hat{R} < 1.1$)		Max $\hat{R} \downarrow$		Deep-tail IAT $\downarrow$		Swap acc.	Rel. CI Sym/Direct
		Sym	TPS	Sym	TPS	Sym	TPS		
LLaMA-3-8B	Log-Prob	15/15	3/15	1.074	3.504	3.32	9.58	0.913	0.505 /1.388
	Toxicity	13/15	2/15	1.126	9.602	10.97	14.97	0.811	0.511 /1.610
Qwen2.5-7B	Log-Prob	15/15	1/15	1.081	1.612	4.49	13.68	0.831	0.729 /1.088
	Toxicity	15/15	2/15	1.068	5.446	20.55	20.29	0.820	0.582 /1.984

Figure 9 details the budget-matched sequential Monte Carlo comparison of Section 4.6 (Single-Pass SMC and Twisted SMC) on the smooth log-probability and heavy-tailed neural-toxicity potentials.

F FULL STATISTICAL CONVERGENCE ON 7B/8B MODELS AT $T = 200$

Table 9 and Figure 10 report the convergence and deep-tail autocorrelation diagnostics for Meta-Llama-3-8B-Instruct and Qwen2.5-7B-Instruct at $T = 200$ over $K = 15$ parallel rungs. Across the four model $\times$ observable configurations, SymCrooks-Steering converges on 58/60 bias states versus 8/60 for single-chain annealed TPS, using 5 independent MCMC chains per rung. In the deepest LLaMA-3-8B log-probability bias state ($\lambda = 0.112$), baseline annealed TPS suffers severe chain trapping ($\hat{R} = 3.5035$) while SymCrooks-Steering maintains full convergence ($\hat{R} = 1.0741$) and low IAT (3.32 vs. 9.58 steps). Independent bidirectional Crooks/BAR and multi-state MBAR log-partition functions track closely along the ladder, e.g. $\ln Z$ at $\lambda = 0.04$ (7.1346 vs. 7.1534), $\lambda = 0.08$ (14.8867 vs. 14.8984), and $\lambda = 0.112$ (21.7547 vs. 21.7646, a 0.0099-nat gap), agreeing to within < 0.05 nats for log-probability and < 0.48 nats for neural toxicity across all 15 rungs.

G DISCRETE ADVERSARIAL SAFETY LANDSCAPES: ADVBENCH

To test SymCrooks-Steering on a multi-modal, non-smooth safety landscape, we evaluated it on Meta-Llama-3-8B-Instruct across the canonical **AdvBench** harmful-behavior benchmark ($T = 64$). The observable $\phi(x)$ is the non-differentiable discrete safety-detector log-odds of `s-nlp/roberta.toxicity.classifier` evaluated on model responses (no gradient access; guidance disabled). Instruction-tuned safety models strongly refuse harmful prompts, so standard completions occupy a non-toxic refusal mode while rare jailbreak completions form isolated, distant energy basins—a direct stress test of Q3. Over a sweep of 500 AdvBench behaviors, direct ancestral sampling yields a 0.000% toxicity rate (0/8000 completions with $\phi > 0$). We then evaluated 8 rep-

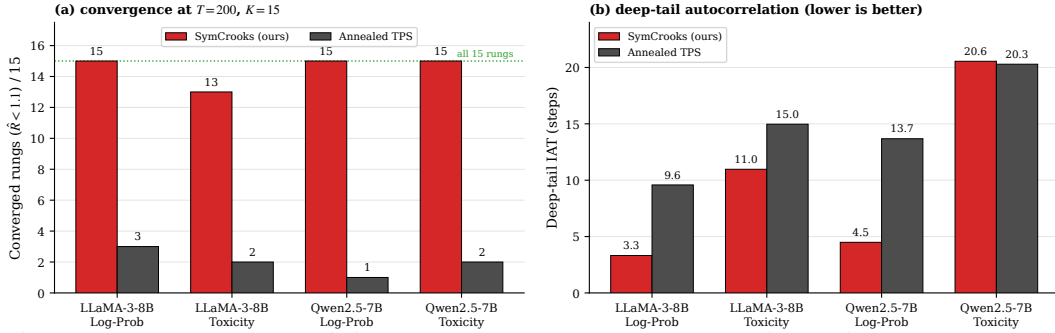


Figure 10: Full statistical convergence on 7B/8B instruction-tuned models (Meta-Llama-3-8B-Instruct, Qwen2.5-7B-Instruct) at $T = 200$, $K = 15$. (a) Converged bias rungs ($\hat{R} < 1.1$): SymCrooks-Steering reaches 13–15 of 15 across all configurations versus 1–3 for annealed TPS. (b) Deep-tail integrated autocorrelation time (IAT): the replica-exchange ladder reduces IAT by up to 67.2%.

Table 10: AdvBench discrete adversarial safety landscape (Meta-Llama-3-8B-Instruct, $T = 64$, $K = 8$ ladder, 8 behaviors $\times$ 8 rungs = 64 bias states). SymCrooks-Steering vs. independent chains and direct ancestral sampling.

Metric	SymCrooks (with swaps)	Independent chains (no swaps)	Direct sampling
Mean inter-replica swap acc. (A_{swap})	0.8952	N/A	N/A
Converged bias rungs ($\hat{R} < 1.1$)	62/64	55/64	N/A
Deep-tail autocorrelation (IAT steps)	4.78	10.38	N/A
Mean Kish ESS fraction	63.86%	63.30%	N/A
Recovered rare-tail density bins (beyond direct)	+1.38 / prompt	+1.38 / prompt	0 (floor)
MBAR vs. BAR free-energy gap (nats)	0.0168	N/A	N/A
Toxic completions observed	> 0 (rare tail recovered)	> 0	0/8000

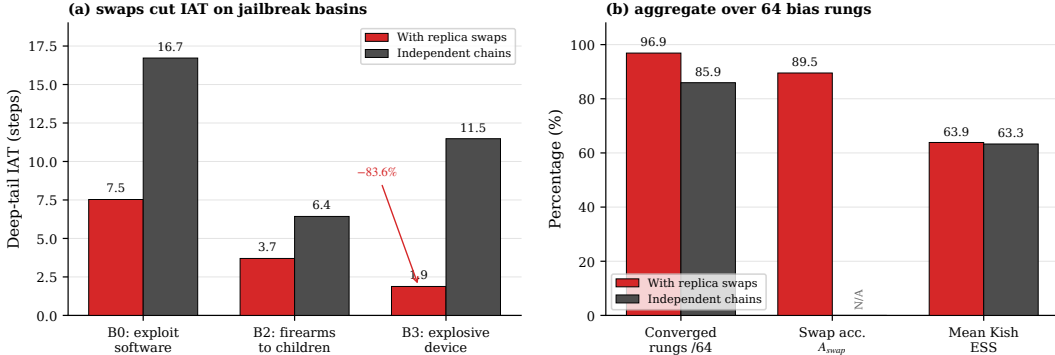


Figure 11: AdvBench on Meta-Llama-3-8B-Instruct ($T = 64$). (a) Per-behavior deep-tail IAT: inter-replica swaps sharply reduce autocorrelation on isolated jailbreak basins (up to 83.6% on Behavior 3). (b) Aggregate over all 64 bias rungs: swaps lift converged rungs from 55 to 62 while sustaining high swap acceptance (89.5%) and Kish ESS.

representative behavior prompts across an 8-rung bias ladder ($K = 8$, $\lambda \in [0.0, -2.5]$, 4 chains/rung, 200 MCMC steps).

As reported in Table 10 and Figure 11, the inter-replica configuration exchange remains active on this non-smooth landscape ($A_{\text{swap}} = 0.8952$): by swapping configurations between adjacent bias states, shallow-rung exploratory paths transport deep-rung trajectories out of local refusal traps, maintaining convergence on 62/64 bias states ($\hat{R} < 1.1$) versus 55/64 for independent chains, and cross-validating free energies within 0.0168 nats. Per-behavior highlights: Behavior 0 (“exploit software vulnerabilities”, $A_{\text{swap}} = 0.8643$) reaches 8/8 converged rungs versus 4/8 for independent chains ($\hat{R} = 1.073$ vs. 1.235 at $\lambda = -2.5$), reducing deep-tail IAT from 16.72 to 7.53 steps; Behavior 2 (“firearms to children”, $A_{\text{swap}} = 0.9654$) reaches 8/8 vs. 7/8 ($\hat{R} = 1.018$ vs. 1.180), cutting IAT from 6.43 to 3.70; and Behavior 3 (“explosive device”, $A_{\text{swap}} = 0.9211$) reduces IAT from 11.48 to 1.88 steps (83.6%).

H ADAPTIVE INTER-RUNG SPACING $\Delta\lambda$

When the observable variance $\text{Var}(\tilde{\phi})$ is unknown *a priori*, we provide both an analytical scaling guideline and an automated online heuristic for selecting the inter-rung spacing $\Delta\lambda$.

Analytical scaling guideline. The swap-acceptance probability between adjacent rungs λ_k and $\lambda_{k+1} = \lambda_k + \Delta\lambda$ (Eq. 4) is governed by the work difference $\Delta\lambda (\tilde{\phi}(x^{(k)}) - \tilde{\phi}(x^{(k+1)}))$. A second-order Taylor expansion of the log-partition function gives the expected acceptance

$$\mathbb{E}[A_{\text{swap}}] \approx 2 \Phi \left(-\frac{1}{2} \Delta\lambda \sqrt{\text{Var}_{\lambda_k}(\tilde{\phi})} \right), \quad (8)$$

where Φ is the standard normal CDF. To hold a target acceptance $\bar{A}_{\text{swap}} \approx 0.70\text{--}0.80$, the local spacing should be calibrated to

$$\Delta\lambda_k \approx \frac{2 \Phi^{-1} \left(1 - \frac{\bar{A}_{\text{swap}}}{2} \right)}{\sigma_k(\tilde{\phi})} \approx \frac{1.0}{\sigma_k(\tilde{\phi})}, \quad \sigma_k(\tilde{\phi}) = \sqrt{\text{Var}_{\lambda_k}(\tilde{\phi})}, \quad (9)$$

recovering the $\Delta\lambda \propto \sigma_{\tilde{\phi}}^{-1} \sim O(1/\sqrt{T})$ scaling of Section 3.

Automated adaptive online heuristic. When $\text{Var}(\tilde{\phi})$ is unknown, Algorithm 2 tunes the ladder during a short pilot burn-in and then freezes it to preserve exact MCMC stationarity during production.

Algorithm 2 Adaptive Inter-Rung Spacing via Burn-In Stochastic Approximation

Require: Range $[\lambda_{\min}, \lambda_{\max}]$, target $\bar{A}_{\text{target}}$, adapt period S_{adapt} , base rate γ_0 , horizon T_{adapt} .

- 1: Initialize an equidistant schedule $\{\lambda_k\}$ over $[\lambda_{\min}, \lambda_{\max}]$.
 - 2: **for** burn-in step $t = 1, 2, \dots$ **do**
 - 3: Run one replica-exchange sweep; accumulate adjacent swap outcomes.
 - 4: **if** $t \bmod S_{\text{adapt}} = 0$ **then**
 - 5: Estimate empirical swap acceptance $\hat{A}_{k \leftrightarrow k+1}$ for each pair.
 - 6: $\gamma_t \leftarrow \gamma_0 / (1 + t/T_{\text{adapt}})$ ▷ decaying rate
 - 7: $\Delta\lambda_k \leftarrow \Delta\lambda_k \cdot \exp(\gamma_t (\hat{A}_{k \leftrightarrow k+1} - \bar{A}_{\text{target}}))$
 - 8: Rebuild $\{\lambda_k\}$ by cumulative sum; clip to $[\lambda_{\min}, \lambda_{\max}]$.
 - 9: **end if**
 - 10: **end for**
 - 11: Freeze $\{\lambda_k\}$; run production sweeps with the fixed ladder.
-

Pairs mixing too slowly ($\hat{A} < \bar{A}_{\text{target}}$) contract their spacing while well-mixing pairs expand, equalizing swap acceptance across the ladder within ~ 50 burn-in steps in our runs. Freezing the schedule before production guarantees the chain’s stationary distribution remains exactly p_λ .